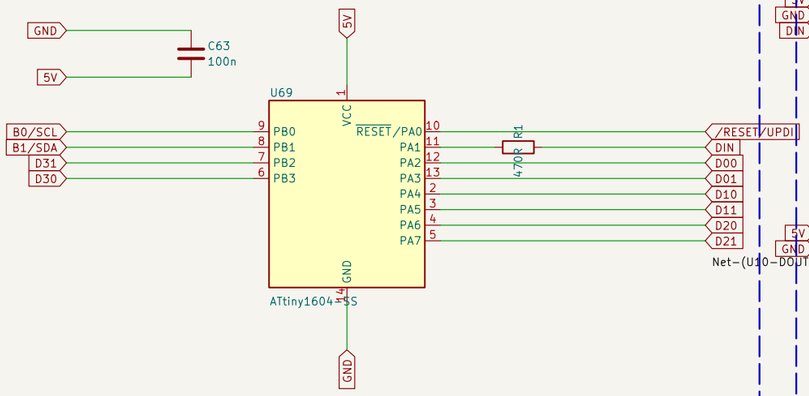
1. MICROCONTROLLER & CONTROLS (ATtiny1604)

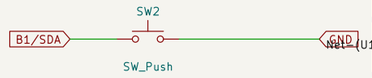
20MHz AVR Core | Dual User Buttons | LED Buffer Resistor



SW1: Brightness Dimmer



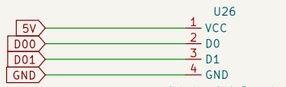
SW2: Mode Selector



2. 4-WAY MAGNETIC SNAP BUS CONNECTORS

Self-Aligning Edge Pogo Connectors with Instant Hardware Detection

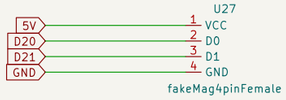
NORTH EDGE (Female)



EAST EDGE (Male)



SOUTH EDGE (Female)



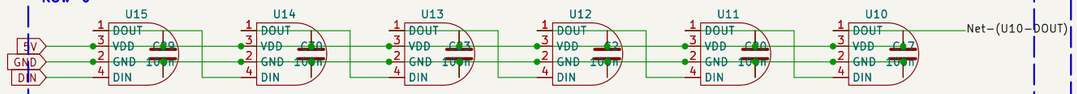
WEST EDGE (Male)



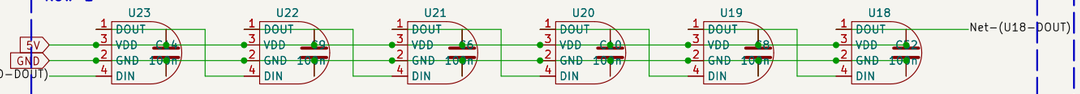
3. 6x6 WS2812D ADDRESSABLE RGB LED MATRIX

36x Diffused LEDs in Serpentine Chain | Local 100n Decoupling Capacitors

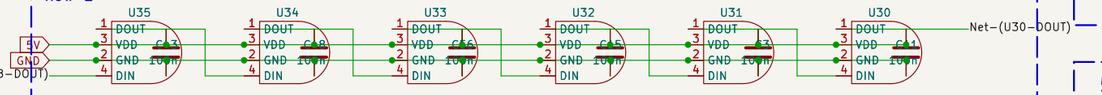
ROW 0



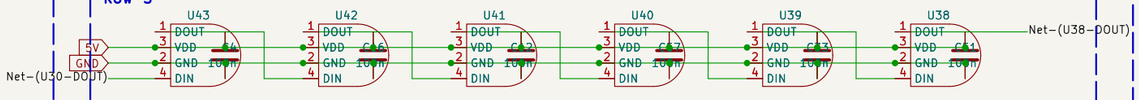
ROW 1



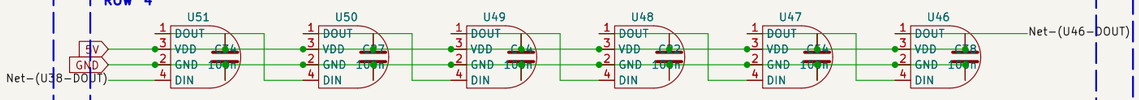
ROW 2



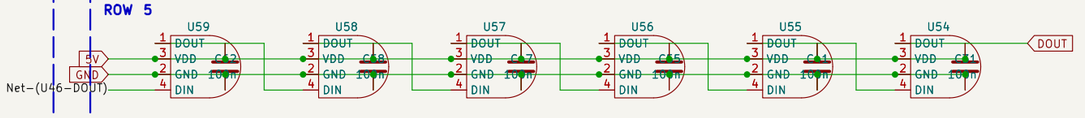
ROW 3



ROW 4



ROW 5

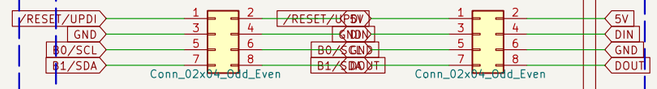


4. PROGRAMMING & EXPANSION HEADERS

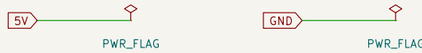
Dual 2x4 SMD Headers for 1-Wire UPDI Flashing & Bus Expansion

J1 (UPDI & EXPANSION)

J2 (MIRROR BUS)



POWER FLAGS & NET DECLARATION



5. THEORY OF OPERATION & SYSTEM NOTES

Architecture, Flashing Guide, and Mechanical Assembly

SYSTEM OVERVIEW:
* Microcontroller: Microchip ATtiny1604-SS (20MHz internal oscillator, 16KB Flash, 1KB SR
* LED Display: 36x WS2812D addressable RGB LEDs arranged in a 6x6 serpentine array.
* Decoupling: Dedicated 100nF ceramic capacitor for each LED and MCU VDD rail.

4-WAY MAGNETIC BUS & AUTODETECT:
* Complementary Gender Architecture:
  - North & South edges: 4-pin Female spring pogo connectors.
  - East & West edges: 4-pin Male target pad connectors.
* Automatic Grid Topology Mapping:
  - Pins D00..D31 form complementary neighbor detection pairs across adjacent tiles.
  - Internal pullups sense tile attachment instantly on contact without software overhead.

FIRMWARE FLASHING (UPDI):
* Single-wire UPDI interface available on Pin 1 of J1/J2 (/RESET/UPDI).
* Compatible with SerialUPDI (CH340/CP2102 + 4.7k resistor + diode) or jtag2updi progra
* Board package: megaTinyCore in Arduino IDE.
* Recommended settings: ATtiny1604, 20MHz internal clock, BOD disabled / 2.6V.

Clean Modular Schematic Layout
Serpentine Daisy Chain with Complementary Neighbor Sensing
6x6 WS2812D Addressable LEDs + ATtiny1604-SS + 4-Way Magnetic Snap Bus
Antigravity Hardware

Sheet: /
File: modularmatrix.kicad_sch

Title: Modular LED Matrix 6x6 (Bitluni Architecture)

Size: A3	Date: 2026-11-01	**Rev: v2.0-aligned**
KiCad E.D.A. 10.0.3		Id: 1/1